

Polarization of Gravitational Waves From Turbulence: Tension Between Simulation and Analytic Approaches

Titus Brock,^{1,*} Murman Gurgenzidze,^{1,†} and Tina Kahniashvili^{1,2,3,‡}

¹*McWilliams Center for Cosmology and Department of Physics,
Carnegie Mellon University, Pittsburgh, PA 15213, USA*

²*School of Natural Sciences and Medicine, Ilia State University, 0194 Tbilisi, Georgia*

³*Abastumani Astrophysical Observatory, Tbilisi, GE-0179, Georgia*

(Dated: May 21, 2026)

CONTENTS

I. Gravitational waves from a turbulent source	1
A. Sourced wave equation	1
B. Green-function solution and the GW spectrum	2
C. Spectral shape from the literature	2
II. Kraichnan–K41 Model (Gogoberidze 2007)	3
A. $\mathbf{k} > \mathbf{k}_0$	3
B. Gogoberidze (2007) Appendix A form	4
C. Mach-dependent peak of the stationary spectrum	4
III. Monochromatic and White-Noise Spectra	4
A. Monochromatic energy spectrum $E(k) = E_0\delta(k - k_0)$	5
1. Kraichnan decorrelation	6
2. Decay model	6
B. Real-space delta-function correlations	6
1. Kraichnan decorrelation	6
2. Decay model	7
Summary of the four dimensionless integrals	7
IV. Numerical results: $f_{\text{peak}}^{\text{GW}}(\tau_*, M_0)$	7
V. Spectrum survey for all derived models	9
References	10

I. GRAVITATIONAL WAVES FROM A TURBULENT SOURCE

Before specializing to particular spectral and temporal models, we collect the standard analytic derivation of the gravitational-wave (GW) spectrum sourced by turbulence [1–5]. This fixes the conventions used throughout and, in particular, shows why the remaining sections reduce the problem to computing the contracted source correlator $H_{ijij}(\mathbf{k}, \omega)$ of Eq. (10) on the diagonal $\omega = k$.

A. Sourced wave equation

Write the tensor perturbations of a spatially flat FLRW background as

$$ds^2 = a^2(\tau)[-d\tau^2 + (\delta_{ij} + h_{ij})dx^i dx^j], \quad \partial_i h_{ij} = h_{ii} = 0, \quad (1)$$

* tbrock@andrew.cmu.edu

† mgurgeni@andrew.cmu.edu

‡ tinatin@andrew.cmu.edu

with h_{ij} transverse-traceless (TT). The linearized Einstein equations give

$$h''_{ij} + 2\frac{a'}{a}h'_{ij} - \nabla^2 h_{ij} = 16\pi G a^2 \Pi_{ij}^{\text{TT}}, \quad (2)$$

where $' = d/d\tau$ and Π_{ij}^{TT} is the TT part of the anisotropic stress. For an incompressible, non-relativistic turbulent fluid the source is the Reynolds stress,

$$\Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau) = \Lambda_{ij,lm}(\hat{k}) T_{lm}(\mathbf{k}, \tau), \quad T_{lm} = w v_l v_m, \quad \Lambda_{ij,lm} = P_{il}P_{jm} - \frac{1}{2}P_{ij}P_{lm}, \quad (3)$$

with $w = \bar{\rho} + \bar{p}$ the enthalpy density, $P_{ij} = \delta_{ij} - \hat{k}_i \hat{k}_j$, and Λ the TT projector. The double TT projection together with the Gaussian (Wick) reduction of the quartic velocity correlator collapses the two-point function of (3) onto the contracted form $R_{ijij} = R_{ii}R_{jj} + \frac{1}{3}R_{ij}R_{ij}$ that appears in Eq. (10).

Turbulence acts on eddy-turnover times short compared with a Hubble time, so over the source lifetime we drop the expansion ($a \approx 1$, friction negligible) and Eq. (2) becomes, in Fourier space, a forced oscillator,

$$h''_{ij}(\mathbf{k}, \tau) + k^2 h_{ij}(\mathbf{k}, \tau) = 16\pi G \Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau). \quad (4)$$

B. Green-function solution and the GW spectrum

The retarded solution of Eq. (4) and its time derivative are

$$h_{ij}(\mathbf{k}, \tau) = \frac{16\pi G}{k} \int_{\tau_i}^{\tau} d\tau' \sin[k(\tau - \tau')] \Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau'), \quad \dot{h}_{ij}(\mathbf{k}, \tau) = 16\pi G \int_{\tau_i}^{\tau} d\tau' \cos[k(\tau - \tau')] \Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau'). \quad (5)$$

With the Isaacson energy density $\rho_{\text{GW}} = \langle \dot{h}_{ij} \dot{h}_{ij} \rangle / (32\pi G)$ and the unequal-time correlator (UETC) of the TT stress,

$$\langle \Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau_1) \Pi_{ij}^{\text{TT}*}(\mathbf{k}', \tau_2) \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}') \Pi(k; \tau_1, \tau_2), \quad (6)$$

inserting (5) and averaging the product $\cos[k(\tau - \tau_1)] \cos[k(\tau - \tau_2)] \rightarrow \frac{1}{2} \cos[k(\tau_1 - \tau_2)]$ over the fast GW oscillations gives the master formula

$$\frac{d\rho_{\text{GW}}}{d \ln k} \simeq \frac{2G k^3}{\pi} \int d\tau_1 \int d\tau_2 \cos[k(\tau_1 - \tau_2)] \Pi(k; \tau_1, \tau_2). \quad (7)$$

The GW spectrum is thus the temporal Fourier transform of the source UETC, evaluated at the on-shell frequency $\omega = k$. For a statistically stationary (or quasi-stationary) source, $\Pi(k; \tau_1, \tau_2) = \Pi(k; \tau_1 - \tau_2)$ and the double time integral factorizes into the source duration T_{src} times the temporal transform $\tilde{\Pi}(k, \omega) = \int d(\Delta\tau) e^{i\omega\Delta\tau} \Pi(k; \Delta\tau)$, which is precisely the spatio-temporal correlator $H_{ijij}(\mathbf{k}, \omega)$ of Eq. (10). Hence

$$\frac{d\rho_{\text{GW}}}{d \ln k} \propto k^3 H_{ijij}(\mathbf{k}, \omega)|_{\omega=k} \implies \Omega_{\text{GW}}(p) \propto p^3 H(p, p), \quad p = \frac{k}{k_0}, \quad q = \frac{\omega}{k_0}, \quad (8)$$

the GW dispersion $\omega = k$ becoming the diagonal $q = p$ used in Sec. V (with characteristic strain $h_c^2(p) = p H(p, p)$). Everything downstream is therefore an evaluation of H_{ijij} , whose spatial part is a convolution of the velocity power spectrum (quadratic source, Eqs. (11)–(12)) and whose temporal part is the decorrelation model carried by $\Pi(k; \tau_1, \tau_2)$.

C. Spectral shape from the literature

For Kolmogorov inertial-range input the analytic results have three robust pieces:

- *Infrared* ($k \ll k_0$): causality forces the stress correlator to be analytic (white) as $k \rightarrow 0$, so $H \rightarrow \text{const}$ and $\Omega_{\text{GW}} \propto k^3$ [3, 4].
- *Peak* ($k \sim k_0$): set by the largest eddies, with parametric amplitude $(d\Omega_{\text{GW}}/d \ln k)_{\text{peak}} \sim (H_*/k_0) \Omega_K^2$, i.e. the squared kinetic-energy fraction suppressed by the source-duration factor $H_*/k_0 \sim H_* \tau_{\text{eddy}}$ [1, 3].
- *Ultraviolet* ($k \gg k_0$): *model-dependent*, fixed by the temporal decorrelation $\Pi(k; \tau_1, \tau_2)$. Kraichnan sweeping gives the Gogoberidze tail [2], while a δ -correlated or a decaying-turbulence kernel give different slopes.

The UV slope is precisely where the four UETC models derived below (δ -spectrum vs. white-noise, Kraichnan vs. decay) differ, which is the focus of the remainder of the paper.

II. KRAICHNAN-K41 MODEL (GOGOBERIDZE 2007)

We use the spatial/temporal Fourier pair $R_{ij}(\mathbf{r}, \tau) = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} F_{ij}(\mathbf{k}, \tau)$ and $f(\eta_k, \tau) = \int \frac{d\omega}{2\pi} e^{-i\omega\tau} \tilde{f}(\eta_k, \omega)$, and adopt the isotropic incompressible model of [2] (Eqs. 23–26),

$$F_{ij}(\mathbf{k}, \tau) = A(k) P_{ij}(\hat{k}) f(\eta_k, \tau), \quad P_{ij}(\hat{k}) = \delta_{ij} - \hat{k}_i \hat{k}_j, \quad (9)$$

with $A(k) = C_k \epsilon^{2/3} k^{-11/3} / 4\pi$, $f(\eta_k, \tau) = \exp(-\frac{\pi}{4} \eta_k^2 \tau^2)$, and $\eta_k = \frac{1}{\sqrt{2\pi}} \epsilon^{1/3} k^{2/3}$.

The fourth-rank Fourier-space correlator $H_{ijkl}(\mathbf{k}, \omega) = (2\pi)^{-4} \int d\tau d^3\xi e^{i\omega\tau - i\mathbf{k}\cdot\xi} R_{ijkl}(\xi, \tau)$ with the contraction $R_{ijij} = R_{ii}R_{jj} + \frac{1}{3}R_{ij}R_{ij}$ (Gogoberidze Eq. 28) reads, in real space,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^4} \int d\tau d^3\mathbf{r} e^{i\omega\tau - i\mathbf{k}\cdot\mathbf{r}} \left[R_{ii}(\mathbf{r}, \tau) R_{jj}(\mathbf{r}, \tau) + \frac{1}{3} R_{ij}(\mathbf{r}, \tau) R_{ij}(\mathbf{r}, \tau) \right]. \quad (10)$$

Inserting the spatial Fourier representation of R_{ij} , the $\mathbf{r}$ integral produces a momentum-conserving $\delta^{(3)}(\mathbf{k} - \mathbf{q} - \mathbf{p})$. Renaming $\mathbf{q} \rightarrow \mathbf{k}_1$, $\mathbf{u} = \mathbf{k} - \mathbf{k}_1$,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^7} \int d\tau e^{i\omega\tau} d^3k_1 \left[F_{ii}(\mathbf{k}_1, \tau) F_{jj}(\mathbf{u}, \tau) + \frac{1}{3} F_{ij}(\mathbf{k}_1, \tau) F_{ij}(\mathbf{u}, \tau) \right]. \quad (11)$$

The temporal Fourier transform then converts the τ integral into a frequency δ -function, leaving the frequency convolution

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^8} \int d^3k_1 d\omega_1 \left[\tilde{F}_{ii}(\mathbf{k}_1, \omega_1) \tilde{F}_{jj}(\mathbf{u}, \omega - \omega_1) + \frac{1}{3} \tilde{F}_{ij}(\mathbf{k}_1, \omega_1) \tilde{F}_{ij}(\mathbf{u}, \omega - \omega_1) \right]. \quad (12)$$

Substituting Eq. (9) and using $P_{ii}(\hat{k}) = 2$ together with $P_{ij}(\hat{k}_1)P_{ij}(\hat{u}) = 1 + (\hat{k}_1 \cdot \hat{u})^2$, the tensor contractions reduce Eq. (12) to

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^8} \int d^3k_1 d\omega_1 A(k_1) A(u) \tilde{f}(\eta_{k_1}, \omega_1) \tilde{f}(\eta_u, \omega - \omega_1) \left[4 + \frac{1}{3} (1 + (\hat{k}_1 \cdot \hat{u})^2) \right]. \quad (13)$$

This is the Fourier representation valid for all k ; we treat $\mathbf{k} > \mathbf{k}_0$ and $\mathbf{k} \leq \mathbf{k}_0$ separately below.

A. $\mathbf{k} > \mathbf{k}_0$

The law of cosines $\hat{k}_1 \cdot \hat{u} = (k^2 - k_1^2 - u^2)/(2k_1u)$ turns the bracket in Eq. (13) into a homogeneous polynomial in (k, k_1, u) ; expanding and grouping gives the geometric kernel

$$\mathcal{K}(k, k_1, u) \equiv \frac{27}{6} + \frac{k^4}{12k_1^2u^2} + \frac{k_1^2}{12u^2} + \frac{u^2}{12k_1^2} - \frac{k^2}{6u^2} - \frac{k^2}{6k_1^2} \quad (\text{cf. Eq. 29 of [2]}). \quad (14)$$

The temporal transform $\tilde{f}(\eta_k, \omega) = (2/\eta_k) \exp(-\omega^2/\pi\eta_k^2)$ combines with $A(k)$ into the single spectral factor

$$g(k, \omega) \equiv A(k) \tilde{f}(\eta_k, \omega) = \frac{2E_k}{4\pi k^2 \eta_k} \exp\left(-\frac{\omega^2}{\pi\eta_k^2}\right), \quad E_k = C_k \epsilon^{2/3} k^{-5/3}. \quad (15)$$

Going to spherical coordinates for $\mathbf{k}_1$ ($d^3k_1 = k_1^2 dk_1 d\phi d\mu$ with $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}_1$), the azimuthal integral contributes 2π . Trading μ for $u = \sqrt{k^2 + k_1^2 - 2kk_1\mu}$ via $d\mu = u du/(kk_1)$, with $u \in [|k - k_1|, k + k_1]$,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^7} \int_0^\infty \frac{k_1 dk_1}{k} \int_{|k-k_1|}^{k+k_1} u du \int_{-\infty}^\infty d\omega_1 g(k_1, \omega_1) g(u, \omega - \omega_1) \mathcal{K}(k, k_1, u). \quad (16)$$

Switching to the dimensionless variables $q = k_1/k$, $s = u/k$, $\Omega = \omega/(\epsilon^{1/3}k^{2/3})$, $\Omega_1 = \omega_1/(\epsilon^{1/3}k^{2/3})$, g becomes $g(k, \omega) = (1/\sqrt{2\pi}) C_k \epsilon^{1/3} k^{-13/3} \exp(-2\Omega^2)$. The product $g(k_1, \omega_1)g(u, \omega - \omega_1)$ thus separates into a dimensional prefactor $\propto k^{-26/3}$ times a manifestly dimensionless Gaussian envelope $\Phi(q, \Omega_1; s, \Omega - \Omega_1) = q^{-13/3} s^{-13/3} \exp[-2\Omega_1^2 q^{-4/3} - 2(\Omega - \Omega_1)^2 s^{-4/3}]$. Collecting all k -dependent factors gives the scaling form

$$H_{ijij}(\mathbf{k}, \omega) = \frac{C_k^2 \epsilon}{256 \pi^8} k^{-5} \mathfrak{H}(\Omega), \quad \mathfrak{H}(\Omega) = \int_0^\infty dq q \int_{-\infty}^\infty d\Omega_1 \int_{|1-q|}^{1+q} ds s \Phi \tilde{\mathcal{K}}(q, s), \quad (17)$$

with $\tilde{\mathcal{K}}(q, s) = \mathcal{K}(k, k_1, u)|_{k_1=qk, u=sk}$ (homogeneous of degree zero). All physical scales are absorbed in the prefactor $C_k^2 \epsilon k^{-5}$.

B. Gogoberidze (2007) Appendix A form

Using the Gaussian-erfc identity (Eq. A.1 of [2], $\int_0^\infty dy e^{-Ay^2 - B(y-x)^2} = \sqrt{\pi/[4(A+B)]} e^{-ABx^2/(A+B)} \operatorname{erfc}[-Bx/\sqrt{A+B}]$), the ω_1 integration in Eq. (16) can be done in closed form, yielding the doubly-bounded form

$$H_{ijij}(k, \omega) = \frac{\varepsilon}{24\pi(2\pi)^{3/2}k} \int_{k_0}^{k_d} dk_1 k_1^{-10/3} \int_{\max(|k_1-k|, k_0)}^{\min(k_1+k, k_d)} du u^{-10/3} (k_1^{-4/3} + u^{-4/3})^{-1/2} \times \left[27 - \frac{k^2}{k_1^2} - \frac{k^2}{u^2} + \frac{k^4}{2k_1^2 u^2} + \frac{k_1^2}{2u^2} + \frac{u^2}{2k_1^2} \right] \exp\left(-\frac{2\varepsilon^{-2/3}\omega^2}{k_1^{4/3} + u^{4/3}}\right) \operatorname{erfc}\left(-\frac{\sqrt{2}\varepsilon^{-1/3}\omega}{(k_1^{-4/3} + u^{-4/3})^{1/2}}\right). \quad (18)$$

k_0 is the outer scale and k_d the dissipation scale; the integration window collapses near the boundaries of the inertial range. This integral is finite as $k \rightarrow 0$.

Define the dimensionless parameters

$$M = \left(\frac{\varepsilon}{k_0}\right)^{1/3}, \quad \frac{k_d}{k_0} = R^{3/4}, \quad p = \frac{k}{k_0}, \quad q = \frac{\omega}{k_0}, \quad x = \left(\frac{k_1}{k_0}\right)^{-4/3}, \quad y = \left(\frac{u}{k_0}\right)^{-4/3}, \quad (19)$$

where $M = (\varepsilon/k_0)^{1/3}$ is the outer-scale turbulent Mach number (sound speed set to unity), equal to the rms eddy velocity u_0 . The Mach number enters the GW kernel through the k -dependent sweeping rate $\eta_k = \frac{1}{\sqrt{2\pi}}\varepsilon^{1/3}k^{2/3} = \frac{M}{\sqrt{2\pi}}k_0^{1/3}k^{2/3}$, which couples frequency to wavenumber in the Gaussian factor of Eq. (18). As a result M sets the *effective* wavenumber support of the integral: the exponential $\exp[-2q^2/(M^2(z^{4/3} + y^{4/3}))]$ (with $z = k_1/k_0$) cuts off contributions with $z^{4/3} + y^{4/3} \lesssim 2q^2/M^2$, so the peak of the GW spectrum shifts with M . The substitution $k_1 = k_0 x^{-3/4}$, $u = k_0 y^{-3/4}$ gives $dk_1 k_1^{-10/3} = -(3/4)k_0^{-7/3}x^{3/4}dx$ and $(k_1^{-4/3} + u^{-4/3})^{-1/2} = k_0^{2/3}(x + y)^{-1/2}$; the integration limits become $x \in [R^{-1}, 1]$ and $y \in [y_{\min}(x), y_{\max}(x)]$ with

$$y_{\max}(x) = \min(|x^{-3/4} - p|^{-4/3}, 1), \quad y_{\min}(x) = \max((x^{-3/4} + p)^{-4/3}, R^{-1}). \quad (20)$$

Plugging into Eq. (18) yields

$$H_{ijij}(p, q) = \frac{3M^3 k_0^{-4}}{256(2\pi)^{3/2}p} \int_{R^{-1}}^1 dx x^{3/4} \int_{y_{\min}}^{y_{\max}} dy y^{3/4} (x + y)^{-1/2} \times \left[27 - p^2(x^{3/2} + y^{3/2}) + \frac{p^4}{2}x^{3/2}y^{3/2} + \frac{1}{2}(x^{-3/2}y^{3/2} + y^{-3/2}x^{3/2}) \right] \times \exp\left(-\frac{2xy}{x+y} \frac{q^2}{M^2}\right) \operatorname{erfc}\left(-\frac{\sqrt{2}q}{M\sqrt{x+y}}\right). \quad (21)$$

C. Mach-dependent peak of the stationary spectrum

Evaluating the kernel (21) on the GW dispersion $\omega = k$ (the diagonal $q = p$ of Eq. (8)) gives the dimensionless energy-density spectrum $\Omega_{\text{GW}}(p) \propto p^3 H_{ijij}(p, p; M, R)$ of stationary Kraichnan-K41 turbulence. The Mach number enters *only* through the sweeping rate $\eta_k \propto M k^{2/3}$, i.e. through the Gaussian $\exp[-\frac{2xy}{x+y} q^2/M^2]$ and the accompanying erfc in Eq. (21). This factor is a soft cutoff that suppresses the integrand once $p \gtrsim M$, so the spectral peak tracks the Mach number, $p_{\text{peak}} \sim M$, until it saturates near the outer scale at large M . Figure 1 shows this directly: the peak slides from $p_{\text{peak}} \simeq 0.05$ at $M = 0.03$ to $p_{\text{peak}} \simeq 2.6$ at $M = 3$, while the amplitude climbs steeply (the M^3 prefactor of Eq. (21) times the widening M -support of the integral). The infrared rise is the universal causal slope $\Omega_{\text{GW}} \propto p^3$ (Sec. IC), common to all M ; only the location and height of the peak respond to M .

The M -dependence of the IR amplitude is fixed entirely by the prefactor: for $p \ll M$ the Gaussian and erfc factors of Eq. (21) reduce to unity, so $H_{ijij}(p \rightarrow 0)$ is set by the M^3 prefactor alone and $\Omega_{\text{GW}} \rightarrow C M^3 p^3$ with C a universal (M -independent) constant. Equivalently the characteristic strain scales as $h_c = \sqrt{pH} \sim M^{3/2} \sqrt{p}$ in the IR. Compensating the spectrum by M^{-3} (the strain by $M^{-3/2}$) therefore collapses every curve onto the single universal p^3 line in the IR, isolating the genuinely M -dependent peak and UV; Fig. 1(b) confirms this collapse numerically (residual spread $< 1.5\%$ at $p \simeq 3 \times 10^{-4}$, and a measured IR exponent $d \ln H / d \ln M \rightarrow 3.00$ as $p \rightarrow 0$).

III. MONOCHROMATIC AND WHITE-NOISE SPECTRA

We now consider two alternative models for the turbulence power spectrum and carry out the same GW-kernel derivation in each case. The first model replaces the Kolmogorov spectrum by a monochromatic (delta-function)

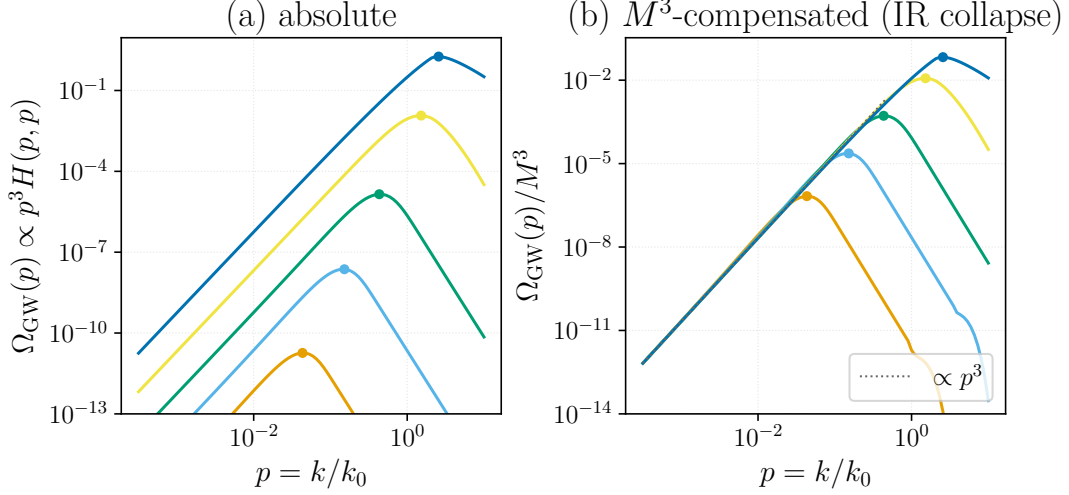


FIG. 1. Stationary Kraichnan-K41 (Gogoberidze 2007) GW spectrum $\Omega_{\text{GW}}(p) \propto p^3 H_{ijij}(p, p; M, R)$, computed from Eq. (21) on the on-shell diagonal of Eq. (8), at $R = 10^4$. The five curves are $M = 0.03, 0.1, 0.3, 1, 3$, ordered by increasing amplitude and peak frequency (lowest, leftmost curve is $M = 0.03$; highest, rightmost is $M = 3$); dots mark each peak. (a) Absolute: shared IR slope p^3 , amplitude growing $\sim M^3$, peak moving right with M ($p_{\text{peak}} \sim M$ in the inertial range, bending over near the outer scale). (b) M^3 -compensated, Ω_{GW}/M^3 : since the IR amplitude scales as M^3 (strain as $M^{3/2}$), all curves collapse onto the universal p^3 line (dotted) in the IR and peel off only at their M -dependent peak. Generated by `Notebooks/stationary_mach_peak.py`.

distribution in wavenumber; the second takes the real-space two-point function to be spatially uncorrelated (delta function in $\mathbf{r}$), which corresponds to a flat (white-noise) spectrum in k -space. For each spatial model we work out both the Kraichnan stationary decorrelation and the decaying-turbulence temporal model, giving four cases in total.

The derivation follows the same sequence of steps as in Sec. II: (i) write H_{ijij} as a double convolution in both $\mathbf{k}$ and ω , (ii) insert the spectral model, (iii) perform the tensor contractions to get $\mathcal{K}$, (iv) switch to spherical coordinates and the variable $u = |\mathbf{k} - \mathbf{k}_1|$, (v) evaluate or simplify the temporal integral, and (vi) introduce dimensionless variables.

A. Monochromatic energy spectrum $E(k) = E_0 \delta(k - k_0)$

Set $E(k) = E_0 \delta(k - k_0)$ so that $A(k) = E_0 \delta(k - k_0)/(4\pi k^2)$ and the spectral tensor reads

$$\tilde{F}_{ij}(\mathbf{k}, \omega) = \frac{E_0 \delta(k - k_0)}{4\pi k^2} P_{ij}(\hat{k}) \tilde{f}(\omega), \quad \int_0^\infty E(k) dk = E_0, \quad (22)$$

with $\tilde{f}(\omega)$ to be specified by the decorrelation model.

Starting from Eq. (13) in spherical coordinates, the radial integral collapses $k_1 \rightarrow k_0$ via $\delta(k_1 - k_0)$, fixing $u = \sqrt{k^2 + k_0^2 - 2kk_0\mu}$. Trading μ for u (azimuth contributes 2π) and using $\delta(u - k_0)$ from $A(u)$ yields the closed-form spatial reduction

$$2\pi \int_{-1}^1 d\mu A(u) = \frac{E_0}{2kk_0^2} \Theta(2k_0 - k), \quad (23)$$

with the Heaviside factor enforcing the triangle inequality $p \equiv k/k_0 \leq 2$. The geometric kernel at $k_1 = u = k_0$ evaluates to

$$\mathcal{K}_0(p) \equiv \mathcal{K}(k, k_0, k_0) = \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12}. \quad (24)$$

Substituting Eqs. (23)–(24) in Eq. (13),

$$H_{ijij}(\mathbf{k}, \omega) = \frac{E_0^2 \Theta(2k_0 - k)}{2(2\pi)^8 k k_0^2} \mathcal{K}_0(p) I_\omega(\omega), \quad I_\omega(\omega) \equiv \int d\omega_1 \tilde{f}(\omega_1) \tilde{f}(\omega - \omega_1). \quad (25)$$

All spatial structure has been resolved in closed form; only the temporal convolution I_ω remains. Because both wavevectors are pinned to k_0 by the two δ -functions, there is no leftover wavenumber integration: the only spatial constraint is the M -independent triangle inequality $\Theta(2-p)$. The Mach number therefore cannot reshape any spatial integration *bounds* here; it enters solely through the temporal decorrelation rate at the fixed scale k_0 , which we tie to M via $\eta_{k_0} = \frac{M}{\sqrt{2\pi}}k_0$ (cf. Eq. (19)). This contrasts with the white-noise case of Sec. III B, where M does enter the surviving wavenumber integral.

1. Kraichnan decorrelation

For $\tilde{f}(\omega) = (2/\eta_0)e^{-\omega^2/\pi\eta_0^2}$ with $\eta_0 \equiv \eta_{k_0} = Mk_0/\sqrt{2\pi}$, the Gaussian convolution gives $I_\omega(\omega) = (2\sqrt{2\pi}/\eta_0)e^{-\omega^2/(2\pi\eta_0^2)}$. With $q \equiv \omega/k_0$ the exponent becomes $\omega^2/(2\pi\eta_0^2) = q^2/M^2$, so Eq. (25) separates into a dimensional prefactor and the boxed dimensionless kernel

$$H_{ijij}(\mathbf{k}, \omega) = \frac{\sqrt{2} E_0^2}{(2\pi)^7 k k_0^2 \eta_0} \mathcal{K}_0(p) \Theta(2-p) e^{-q^2/M^2}, \quad \boxed{\mathfrak{H}_{\text{Kraichnan}}^{(\delta k)}(p, q; M) = \frac{\mathcal{K}_0(p)}{p} \Theta(2-p) e^{-q^2/M^2}}. \quad (26)$$

There is no remaining integral. The p -shape is fixed; M acts only as the Gaussian width in q , so the GW spectrum peaks at $q \sim M$.

2. Decay model

For decaying turbulence, $\tilde{f}(\omega) = \hat{g}(\omega\tau_1)$ with $\hat{g}(\bar{q}) = e^{i\bar{q}}(-i\bar{q})^{-5/3}\Gamma(\frac{1}{3}, -i\bar{q})$ and $\tau_1 = 1/\eta_0 = \sqrt{2\pi}/(Mk_0)$ the decay time at scale k_0 . With $q = \omega/k_0$ the natural temporal argument is $\bar{q} \equiv \omega\tau_1 = \sqrt{2\pi}q/M$; changing variables in Eq. (25), the temporal factor becomes $I_\omega = (1/\tau_1) \int dq_1 \hat{g}(q_1) \hat{g}(\bar{q} - q_1)$, which has no closed form. The dimensionless result is

$$\boxed{\mathfrak{H}_{\text{decay}}^{(\delta k)}(p, q; M) = \frac{\mathcal{K}_0(p)}{p} \Theta(2-p) \int_{-\infty}^{\infty} dq_1 \hat{g}(q_1) \hat{g}(\bar{q} - q_1), \quad \bar{q} = \frac{\sqrt{2\pi} q}{M}}. \quad (27)$$

The spatial part is fully resolved by the two δ -functions; only a 1-D frequency integral remains, and M enters it solely through the rescaled frequency $\bar{q}$.

B. Real-space delta-function correlations

Take the spatial part of R_{ij} to be a $\delta^{(3)}(\mathbf{r})$, with velocity variance v_0^2 and temporal decorrelation $T(\eta_k, \tau)$:

$$R_{ij}(\mathbf{r}, \tau) = v_0^2 \delta^{(3)}(\mathbf{r}) P_{ij} T(\eta_k, \tau), \quad \tilde{F}_{ij}(\mathbf{k}, \omega) = v_0^2 P_{ij}(\hat{k}) \tilde{T}(\eta_k, \omega), \quad (28)$$

so $A(k) = v_0^2$ is k -independent (i.e. $E(k) \propto k^2$, ultraviolet-dominated); the integrals must be cut off between k_0 and k_d . Crucially, we adopt the same k -dependent sweeping rate as in Sec. II, $\eta_k = \frac{1}{\sqrt{2\pi}}\varepsilon^{1/3}k^{2/3} = \frac{M}{\sqrt{2\pi}}k_0^{1/3}k^{2/3}$, so that $\tilde{T}$ depends on the wavenumber and the temporal factor no longer decouples from the spatial integral. With $A(k_1) = A(u) = v_0^2$ in Eq. (11) and dimensionless variables $z = k_1/k_0$, $y = u/k_0$, $p = k/k_0$, $q = \omega/k_0$, $R^{3/4} = k_d/k_0$,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{v_0^4 k_0^2}{(2\pi)^7} \int_1^{R^{3/4}} \frac{z dz}{p} \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{\mathcal{K}}(p, z, y) J(\omega; k_1, u), \quad J(\omega; k_1, u) \equiv \int d\omega_1 \tilde{T}(\eta_{k_1}, \omega_1) \tilde{T}(\eta_u, \omega - \omega_1), \quad (29)$$

where $\tilde{\mathcal{K}}(p, z, y) = \mathcal{K}(k, k_1, u)|_{k_1=z k_0, u=y k_0}$ and the frequency factor J now carries the k_1, u dependence inherited from η_{k_1}, η_u .

1. Kraichnan decorrelation

With $\tilde{T}(\eta_k, \omega) = (2/\eta_k)e^{-\omega^2/\pi\eta_k^2}$ the convolution combines two Gaussians of *different* widths η_{k_1}, η_u ; using $\int d\omega_1 e^{-a\omega_1^2} e^{-b(\omega-\omega_1)^2} = \sqrt{\pi/(a+b)} e^{-ab\omega^2/(a+b)}$ with $a = 1/\pi\eta_{k_1}^2$, $b = 1/\pi\eta_u^2$, and $\eta_{k_1}^2 + \eta_u^2 = \frac{M^2}{2\pi}k_0^{2/3}(k_1^{4/3} + u^{4/3})$,

the frequency factor becomes

$$J(\omega; k_1, u) = \frac{4\pi\sqrt{2\pi}}{Mk_0^{1/3}\sqrt{k_1^{4/3} + u^{4/3}}} \exp\left(-\frac{2\omega^2}{M^2k_0^{2/3}(k_1^{4/3} + u^{4/3})}\right). \quad (30)$$

Inserting this into Eq. (29) and absorbing the constants into the dimensional prefactor gives the boxed dimensionless kernel

$$\mathfrak{H}_{\text{Kraichnan}}^{(\delta r)}(p, q; M, R) = \int_1^{R^{3/4}} \frac{z dz}{p} \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{\mathcal{K}}(p, z, y) (z^{4/3} + y^{4/3})^{-1/2} e^{-2q^2/[M^2(z^{4/3} + y^{4/3})]}. \quad (31)$$

The exponential now acts as an M -dependent effective cutoff on the wavenumber integration: contributions require $z^{4/3} + y^{4/3} \gtrsim 2q^2/M^2$, so as M grows the effective wavenumber support widens and the GW spectrum peak shifts. This is precisely the structure of the Gogoberidze exponent $\exp[-2\varepsilon^{-2/3}\omega^2/(k_1^{4/3} + u^{4/3})]$ in Eq. (18); the k -dependent sweeping rate, rather than a fixed η_0 , is what makes M control the integration support.

2. Decay model

For decaying turbulence $\tilde{T}(\eta_k, \omega) = \hat{g}(\omega\tau_k)$ with the same $\hat{g}(q) = e^{iq}(-iq)^{-5/3}\Gamma(\frac{1}{3}, -iq)$ as Sec. III A 2, but now with the k -dependent decay time $\tau_k = 1/\eta_k = \sqrt{2\pi}/(Mk_0^{1/3}k^{2/3})$. The temporal factor is the convolution $J(\omega; k_1, u) = \int d\omega_1 \hat{g}(\omega_1\tau_{k_1})\hat{g}((\omega - \omega_1)\tau_u)$, which has no closed form and, because τ_{k_1}, τ_u depend on the wavenumbers, must remain *inside* the spatial integral. Writing $\tilde{J}(q; z, y, M)$ for this convolution with the per-mode dimensionless frequencies fixed by $\tau_{k_1}, \tau_u \propto 1/M$, the boxed result is

$$\mathfrak{H}_{\text{decay}}^{(\delta r)}(p, q; M, R) = \int_1^{R^{3/4}} \frac{z dz}{p} \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{\mathcal{K}}(p, z, y) \tilde{J}(q; z, y, M). \quad (32)$$

The decay rates scale as $\eta_k \propto Mk_0^{1/3}k^{2/3}$, so M again sets the effective wavenumber support. Unlike the fixed- η_0 case, the three integrals no longer factorize: the k -dependence of the decay rate couples the frequency convolution to the wavenumber integration, mirroring the behavior of Eq. (18).

Summary of the four dimensionless integrals

Spectrum	Decorrelation	Dimensionless integral
$E_0\delta(k - k_0)$	Kraichnan	Eq. (26): closed form; M in frequency scale only
$E_0\delta(k - k_0)$	Decay	Eq. (27): 1D frequency integral; M in $\bar{q}$
$\delta^{(3)}(\mathbf{r})$	Kraichnan	Eq. (31): 2D wavevector integral; M -cutoff in bounds
$\delta^{(3)}(\mathbf{r})$	Decay	Eq. (32): 2D wavevector $\times$ nested 1D frequency; M -cutoff

IV. NUMERICAL RESULTS: $f_{\text{peak}}^{\text{GW}}(\tau_*, M_0)$

We scan the peak GW frequency $f_{\text{peak}}^{\text{GW}} = \text{argmax}_{\omega}[\omega\Omega_{\text{GW}}(\omega)]/2\pi$, with Ω_{GW} from the master spectrum (8), over the (τ_*, M_0) plane for a delta-shell decaying source evolved with the BK2016 self-similar closure and time-integrated over the emission window (Notebooks/f_peak_vs_tau_mach.py). Two temporal-correlation kernels are compared through their dimensionless transform $\mathcal{G}(q) = \widehat{R^2}(q)$: the random-phase BK2016 form (Model A) and the coherent eddy (Model B), shown in Fig. 2.

Across all four self-similar classes the ratio $f_{\text{peak}}^A/f_{\text{peak}}^B$ is *independent of* (τ_*, M_0) : the self-similar closure makes both the eddy frequency ω_e and the sweeping rate η_0 scale as $(1 + T/\tau_*)^{-1}$, so the two kernels inherit the same overall frequency scaling and differ only in their fixed dimensionless peak position. The model difference therefore collapses to a single per-class constant (Fig. 3): $f_A/f_B \simeq 0.37, 0.36$ for the HD (Loitsiansky, Saffman) classes and 0.83, 0.78 for the MHD (nonhelical, helical) classes – Model A always peaks lower. The large HD offset comes from the side-peak

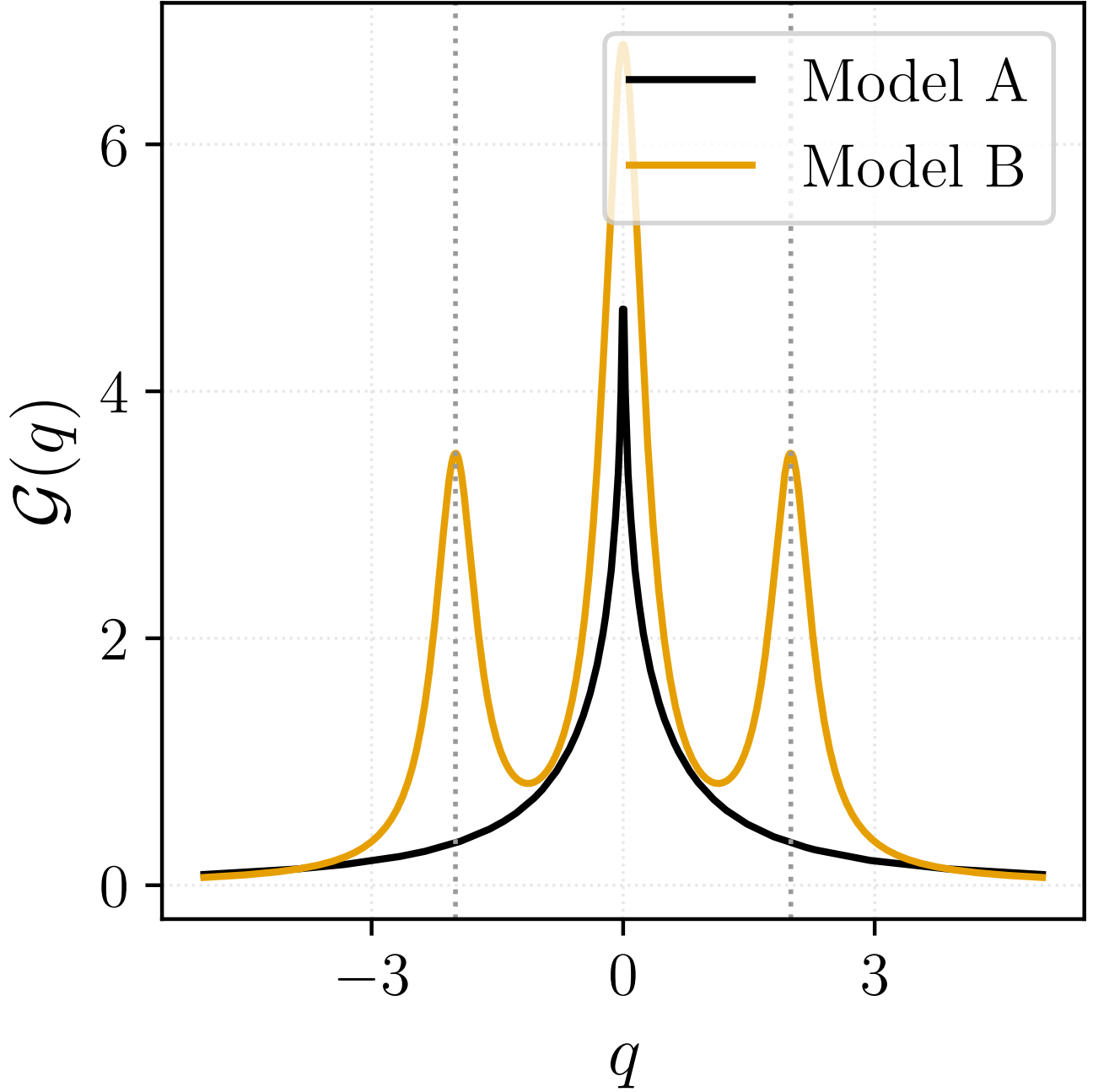


FIG. 2. Dimensionless temporal kernel $\mathcal{G}(q) = \widehat{R^2}(q)$ for the two correlation models considered in Sec. IV. Model A uses the random-phase BK2016 form $R_A(\sigma) = (1 + |\sigma|)^{-2/3}$; Model B is the coherent eddy $R_B(\sigma) = \cos(\sigma) e^{-\alpha|\sigma|}$ with $\alpha = 0.15$.

of the coherent kernel at the eddy frequency ($q \simeq \pm 2$, the $\cos \sigma$ oscillation of R_B), which wins the $\omega \Omega_{\text{GW}}$ argmax for the steep-decay HD classes but nearly coincides with the broad Model-A peak for MHD. Because the per-model maps differ only by this constant, we display the ratio rather than the eight individual heatmaps.

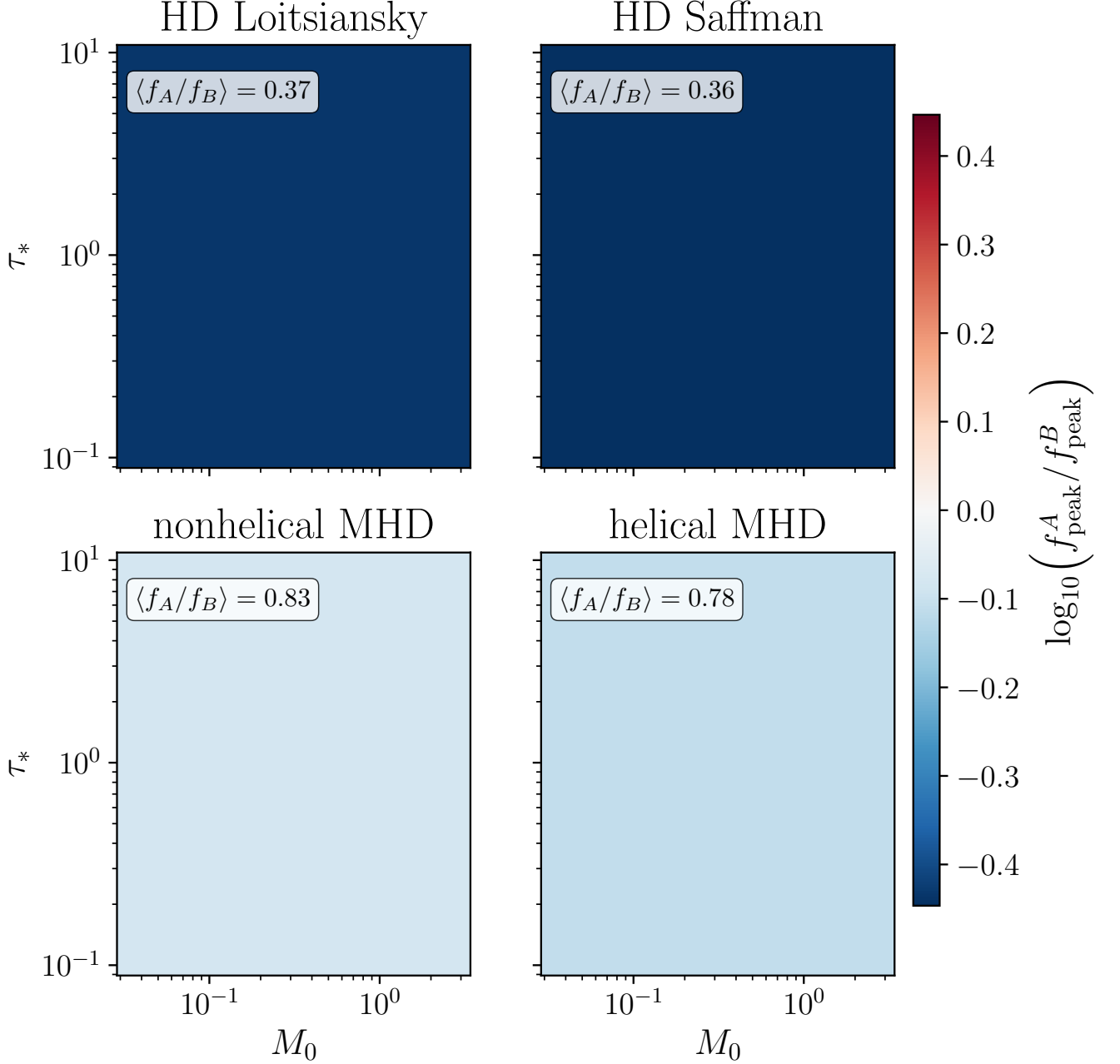


FIG. 3. Model A/B ratio of the peak GW frequency, $\log_{10}(f_{\text{peak}}^A/f_{\text{peak}}^B)$, on the (τ_*, M_0) plane for the four BK2016 self-similar decay classes (HD Loitsiansky $q = 2/7$, $\beta = 4$; HD Saffman $q = 1/3$, $\beta = 3$; nonhelical MHD $q = 1/2$, $\beta = 1$; helical MHD $q = 2/3$, $\beta = 0$). Each panel is flat: the ratio is independent of (τ_*, M_0) , fixed only by the kernel shapes, with the per-class value $\langle f_A/f_B \rangle$ annotated. Model A (random-phase) peaks below Model B (coherent eddy) everywhere; the HD classes show the larger offset.

V. SPECTRUM SURVEY FOR ALL DERIVED MODELS

In this section we plot, on a common convention, the dimensionless GW energy-density spectrum

$$\Omega_{\text{GW}}(p) \propto p^3 H(p, p), \quad p = k_{\text{GW}}/k_0, \quad (33)$$

along the sound-cone diagonal $q = p$ for every model derived in this paper. The p^2 factor relative to the characteristic strain $h_c^2(p) = p H(p, p)$ is the phase-space measure d^3k ; together with causality and a finite white-noise

source it predicts $\Omega_{\text{GW}} \sim p^3$ in the IR, followed by a peak near $p \sim 1$ and a decay in the UV. A dotted grey line shows the p^3 reference; cases whose IR slope deviates from 3 by more than ± 0.5 are flagged in the build log of `Notebooks/all_spectra.py`. Models with a $1/p$ pole at the origin (monochromatic $E_0 \delta(k - k_0)$ source) cannot satisfy the universal scaling because the fluid itself carries no IR power.

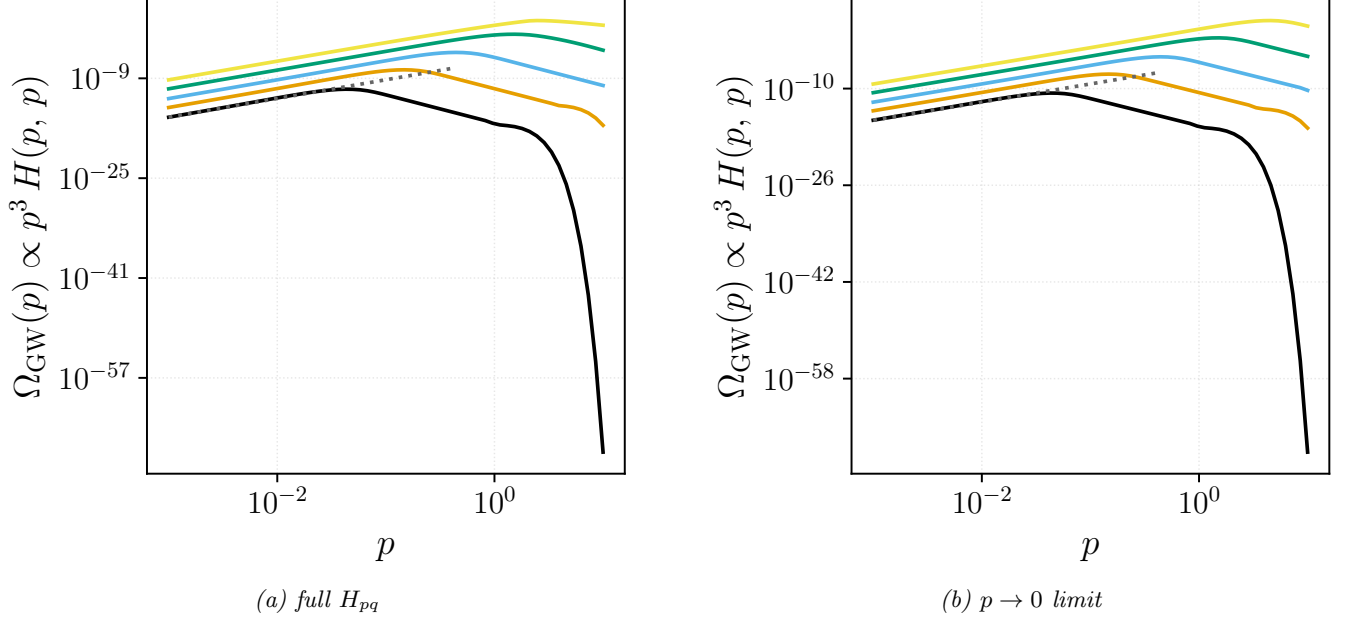


FIG. 4. Kraichnan decorrelation + Kolmogorov spectrum. Five curves per panel correspond to $M = 0.03, 0.1, 0.3, 1.0, 3.0$ at $R = 10^4$. Panel (a) is the full numerical H_{pq} of Eq. (21) (Sec. II); panel (b) is the aeroacoustic $p \rightarrow 0$ limit (Kahniashvili et al. 2008, [3]). Both give the universal IR slope $\Omega_{\text{GW}} \propto p^3$.

-
- [1] A. Kosowsky, A. Mack, and T. Kahniashvili, *Phys. Rev. D* **66**, 024030 (2002), [arXiv:astro-ph/0111483 \[astro-ph\]](#).
 - [2] G. Gogoberidze, T. Kahniashvili, and A. Kosowsky, *Phys. Rev. D* **76**, 083002 (2007), [arXiv:0705.1733 \[astro-ph\]](#).
 - [3] T. Kahniashvili, L. Campanelli, G. Gogoberidze, Y. Maravin, and B. Ratra, *Phys. Rev. D* **78**, 123006 (2008), [Erratum: *Phys. Rev. D* **79**, 109901 (2009)], [arXiv:0809.1899 \[astro-ph\]](#).
 - [4] C. Caprini and D. G. Figueroa, *Class. Quant. Grav.* **35**, 163001 (2018), [arXiv:1801.04268 \[astro-ph.CO\]](#).
 - [5] M. Maggiore, *Gravitational Waves. Vol. 1: Theory and Experiments* (Oxford University Press, 2007).

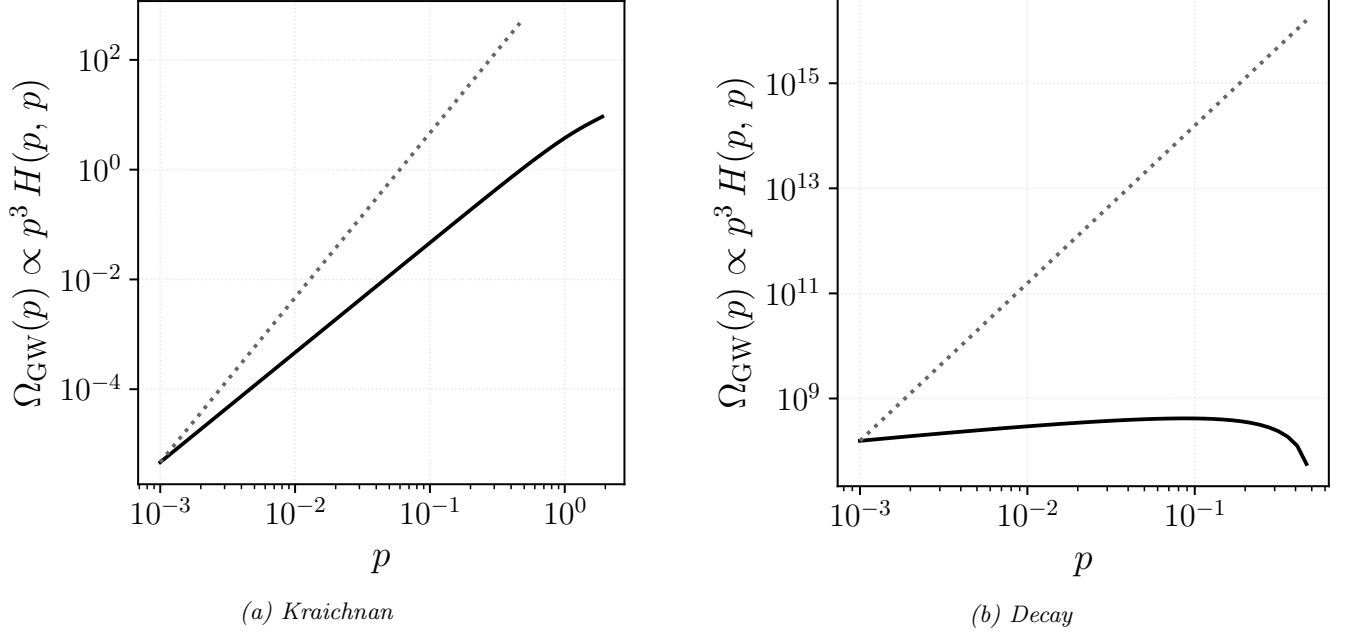


FIG. 5. Monochromatic spectrum $E(k) = E_0 \delta(k - k_0)$ (no free parameters at the dimensionless level). Panel (a): Kraichnan decorrelation, closed form of Eq. (26), $\Omega_{\text{GW}}(p) \propto p^3 K_0(p)/p \Theta(2 - p) e^{-p^2/(2\pi)}$ with IR slope **2** (flagged: $1/p$ pole from the monochromatic source removes one power of p from the universal scaling). Panel (b): decay temporal convolution of Eq. (27), IR slope **0.3** (flagged: the temporal-tail extra suppression further reduces the IR power).

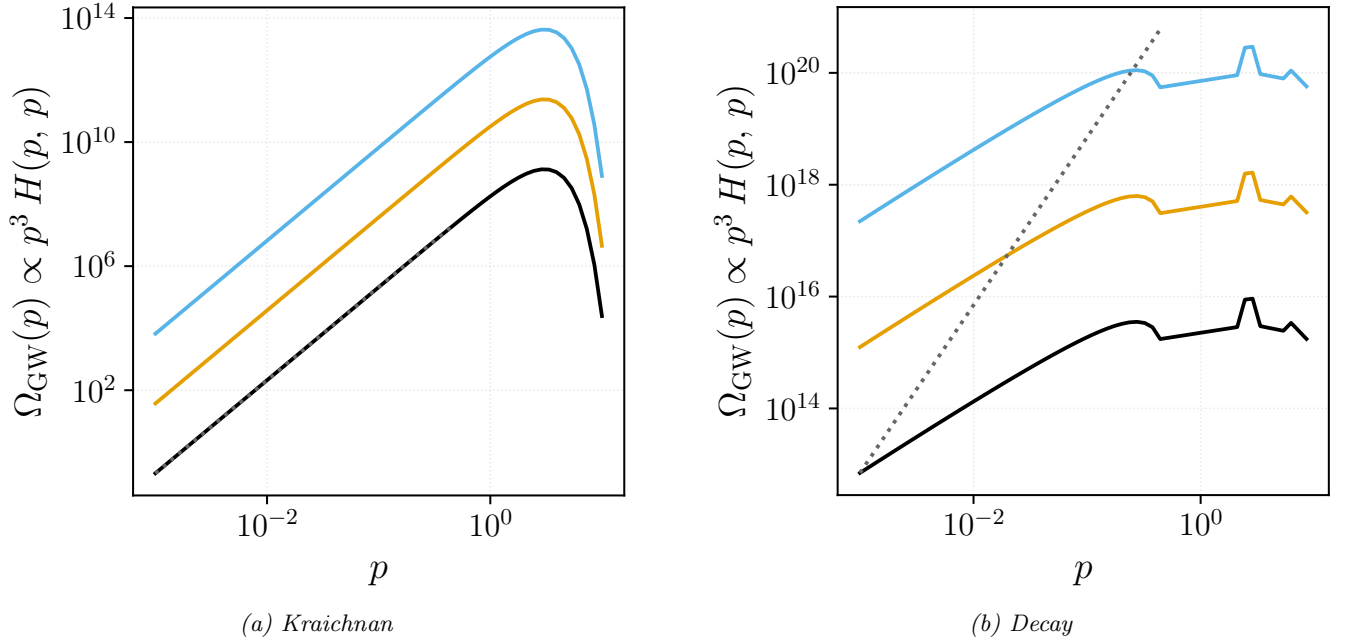


FIG. 6. White-noise real-space correlations $R_{ij} \propto \delta^{(3)}(\mathbf{r})$. Three curves per panel correspond to $R = 10^3, 10^4, 10^5$. Panel (a): Kraichnan decorrelation of Eq. (31), matches the universal IR slope **3**. Panel (b): decay temporal model of Eq. (32), IR slope **1.3** (flagged: temporal-convolution tails pull power down at small p).